Mathematics

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September 4, 2025

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1 Fundamentals of Higher Mathematics

Generally covers four areas: single-variable calculus, linear algebra, multivariable calculus, and differential equations. As this is fundamentals, the focus is generally on computation instead of proofs.

1.1 Single Variable Calculus

Calculus is the branch of mathematics that studies change and accumulation. It has two main branches:

i. Differential Calculus: Concerned with the rate at which things change (e.g. velocity).

ii. Integral Calculus: Concerned with accumulation (e.g. area under a curve).

Calculus underpins physics, engineering, economics, biology, and more.

Definition 1.1.1. The *derivative* is defined as:

$$\lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0)$$

Definition 1.1.2. A function $f(x)$ is continuous at x_0 when

$$\lim_{x \to x_0} f(x) = f(x_0)$$

Definition 1.1.3. The *right-hand* and *left-hand* limits are as follows:

i. *Right-Hand Limit*: $\lim_{x \to x_0^+} f(x)$ means $\lim_{x \to x_0} f(x)$ for $x > x_0$

ii. *Left-Hand Limit*: $\lim_{x \to x_0^-} f(x)$ means $\lim_{x \to x_0} f(x)$ for $x < x_0$

Definition 1.1.4. A *removable discontinuity* is where

$$\lim_{x \to x_0^+} f(x) = \lim_{x \to x_0^-} f(x)$$

but not at $f(x_0)$, or if $f(x_0)$ is undefined.

Definition 1.1.5. A *jump discontinuity* is where $\lim_{x \to x_0^+} f(x)$ and $\lim_{x \to x_0^-} f(x)$ exists, but not equal.

Definition 1.1.6. A *infinite discontinuity* is where $\lim_{x \to x_0^+} f(x) \neq \lim_{x \to x_0^-} f(x) \in \{+\infty, -\infty\}$

Theorem 1.1.7. *Differentiability Implies Continuity*
If f is differentiable at x_0, then f is continuous at x_0.

Proof.

$$\lim_{x \to x_0} (f(x) - f(x_0)) = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} \right] (x - x_0) = f'(x_0) \cdot 0 = 0$$

□

Definition 1.1.8. *Product Formula*
The derivative of product of two functions.

$$(u \cdot v)' = u' \cdot v + u \cdot v'$$

Definition 1.1.9. *Quotient Formula*
The derivate of ratio of two differentiable functions. Let $v(x) \neq 0$, then

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Definition 1.1.10. *Chain Rule*
The derivative of composition of two functions.

$$\frac{d}{dt} f(g(t)) = f'(g(t))g'(t)$$

May be used to differentiate implicitly defined functions.

Theorem 1.1.11. *Taylor's Theorem*
Let $k \geq 1$ be an integer and let function $f : \mathbb{R} \to \mathbb{R}$ be k-times differentiable at point $a \in \mathbb{R}$. Then there exists a function $h_k : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) = \sum_{i=0}^{k} \frac{f^{(i)}(a)}{i!}(x-a)^i + h_k(x)(x-a)^k$$

where $\lim_{x \to a} h_k(x) = 0$ is the Peano form of the remainder.

Theorem 1.1.12. *Mean Value Theorem*
Let $f : [a, b] \to \mathbb{R}$ be a continuous function on closed interval $[a, b]$ and differentiable on open interval (a, b) where $a < b$. Then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem 1.1.13. *First Fundamental Theorem of Calculus*
Let f be continuous real-valued function on closed interval $[a, b]$ Let F be function defined for all $x \in [a, b]$, by

$$F(x) = \int_a^x f(t)dt$$

Then F is uniformly continuous on $[a, b]$ and differentiable on open interval (a, b), and

$$F'(x) = f(x)$$

for all x in (a, b) so F is an antiderivative of f.

Remark 1.1.14. *Change of Variable*
If $f(x) = g(u(x))$, then $du = u'(x)dx$, hence

$$\int g(u)du = \int g(u(x))u'(x)dx = \int f(x)u'(x)dx$$

For definite integrals,

$$\int_{x_1}^{x_2} f(x)u'(x)dx = \int_{u_1}^{u_2} g(u)d(u)$$

where $u_1 = u(x_1), u_2 = u(x_2)$.

Theorem 1.1.15. *Second Fundamental Theorem of Calculus*
Let $f : \mathbb{R} \to \mathbb{R}$ on $[a, b]$ and F a continuous function on $[a, b]$ which is an antiderivative of f in (a, b),

$$F'(x) = f(x)$$

If f is Riemann integrable on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Remark 1.1.16. *Integration by Trigonometric Substitution*
Evaluate trigonometric integral, work backwards to original variable by using $\text{trig}(\text{trig}^{-1})$ with right-triangle.

i. $a^2 - x^2$ use $x = a \sin u$ and $dx = a \cos u \; du$
ii. $a^2 + x^2$ use $x = a \tan u$ and $dx = a \sec^2 u \; du$
iii. $x^2 - a^2$ use $x = a \sec u$ and $dx = a \sec u \tan u \; du$

Remark 1.1.17. *Systematic Integration of Rational Functions*
For a rational function P/Q (ratio of polynomials), by method of partial fractions, factor Q completely into linear and quadratic factors, and write P/Q as a sum of simpler terms.

i. If $Q(x) = (x-a)^m(x-b)^n$, then the expression is

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_m}{(x-a)^m} + \frac{B_1}{x-b} + \frac{B_2}{(x-b)^2} + \cdots + \frac{B_n}{(x-b)^n}$$

ii. If there are quadratic factor such as $(Ax^2+Bx+C)^p$, then there are terms

$$\frac{a_1x+b_1}{Ax^2+Bx+C}+\frac{a_2x+b_2}{(Ax^2+Bx+C)^2}+\cdots+\frac{a_px+b_p}{(Ax^2+Bx+C)^p}$$

for each factor. To integrate, complete the square and make trigonometric substitution.

Remark 1.1.18. *Integration by Parts*

$$\int_b^a uv'\,dx = uv|_a^b - \int_a^b u'v\,dx$$

This is used when $u'v$ is simpler than uv' (often the case when u' is simpler than u).

Remark 1.1.19. *Arc-Length*
Given $ds=\sqrt{dx^2+dy^2}$, depending on whether to integrate with respect to x,t,y, this is

$$ds=\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx;\quad ds=\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt;\quad ds=\sqrt{\left(\frac{dx}{dy}\right)^2+1}\,dy$$

Remark 1.1.20. *Surface Area for Surface of Revolution*

i. Around x-axis: (requires formula for $y=y(x)$)

$$2\pi y\,ds = 2\pi y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

ii. Around y-axis: (requires formula for $x=x(y)$)

$$2\pi x\,ds = 2\pi x\sqrt{\left(\frac{dx}{dy}\right)^2+1}\,dy$$

Remark 1.1.21. *Polar Coordinates*
Given $x=r\cos\theta$, $y=r\sin\theta$ (or more rarely $r=\sqrt{x^2+y^2}\theta$, $\theta=\tan^{-1}(y/x)$)

i. Find the polar equation for a curve from its equation in (x,y) variables by substitution.
ii. Sketch curves given in polar coordinates and find range of θ for integration.
iii. Integration area formula is then

$$\int_{\theta_1}^{\theta_2}\frac{1}{2}r^2\,d\theta$$

Theorem 1.1.22. *L'Hopital Rule*
For functions f and g defined on open interval I and differentiable on $I\backslash\{c\}$ for a accumulation point c of I, if

$$\lim_{x\to c}f(x)=\lim_{x\to c}g(x)=0\quad or\quad \pm\infty$$

and $g'(x)\neq 0$ for all $x\in I\backslash\{c\}$, and $\lim_{x\to c}\frac{f'(x)}{g'(x)}$ exists, then

$$\lim_{x\to c}\frac{f(x)}{g(x)}=\lim_{x\to c}\frac{f'(x)}{g'(x)}$$

1.2 Linear Algebra

Linear algebra is the branch of mathematics that studies vectors, matrices, and linear transformations. It provides the foundation for solving systems of equations and analysing multidimensional data. Its applications span across machine learning, computer graphics, physics, finance, and engineering.
References from Garcia and Horn (2017) and Axler (2024).

1.2.1 Vector Spaces

Definition 1.2.1. *Vector Space*
Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Then $\mathcal{V}$ is a vector space over a field $\mathbb{F}$ if the following are satisfied:

i. Identities: $\exists\ \mathbf{0} \in \mathcal{V}$ such that $\mathbf{0} + \mathbf{u} = \mathbf{u}\ \forall\ \mathbf{u} \in \mathcal{V}$, and $\exists\ \mathbf{1} \in \mathcal{V}$ such that $\mathbf{1u} = \mathbf{u}\ \forall\ \mathbf{u} \in \mathcal{V}$
ii. Commutativity: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}\ \forall\ \mathbf{u}, \mathbf{v} \in \mathcal{V}$
iii. Associativity: $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ and $(ab)\mathbf{u} = a(b\mathbf{u})\ \forall\ a, b \in \mathbb{F}$ and $\forall\ \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$
iv. Distributive: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ and $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}\ \forall\ a, b \in \mathbb{F}$ and $\forall\ \mathbf{u}, \mathbf{v} \in \mathcal{V}$
v. Additive Inverse: $\forall\ \mathbf{u} \in \mathcal{V}\ \exists$ unique vector $-\mathbf{u} \in \mathcal{V}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
vi. Multiplicative Inverse: $\forall\ \mathbf{u} \in \mathcal{V}$ with $\mathbf{u} \neq \mathbf{0}\ \exists$ unique $\mathbf{w} \in \mathcal{V}$ such that $\mathbf{uw} = 1$

Remark 1.2.2. *Properties of Vector Spaces*
Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$.

i. A vector space has a unique additive identity
ii. Every element in a vector space has a unique additive inverse
iii. $\mathbf{0v} = \mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$
iv. $a\mathbf{0} = \mathbf{0}$ for every $a \in \mathbb{F}$
v. $(-1)\mathbf{v} = -\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$

Definition 1.2.3. *Subspace*
A subspace of an $\mathbb{F}$-vector space $\mathcal{V}$ is a subset $\mathcal{U}$ of $\mathcal{V}$ that is an $\mathbb{F}$-vector space with the same vector addition and scalar multiplication operations as in $\mathcal{V}$. To check $\mathcal{U}$ satisfies the following:

i. Additive Identity: $\mathbf{0} \in \mathcal{U}$
ii. Closure Under Addition: $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ implies $\mathbf{u} + \mathbf{v} \in \mathcal{U}$
iii. Closed Under Scalar Multiplication: $a \in \mathbb{F}$ and $\mathbf{u} \in \mathcal{U}$ implies $a\mathbf{u} \in \mathcal{U}$.

Definition 1.2.4. *Sum of Subspaces*
Suppose $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. The sum of $\mathcal{U}_1, \ldots, \mathcal{U}_m$ as denoted by $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is defined to be set of all possible sums of elements of $\mathcal{U}_1, \ldots, \mathcal{U}_m$:

$$\mathcal{U}_1 + \cdots + \mathcal{U}_m \coloneqq \{\mathbf{u}_1 + \cdots + \mathbf{u}_m : \mathbf{u}_1 \in \mathcal{U}_1, \ldots, \mathbf{u}_m \in \mathcal{U}_m\}$$

Definition 1.2.5. *Direct Sum of Subspaces*
If $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$ such that $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, then every element of $\mathcal{V}$ may be written as

$$\mathbf{u}_1 + \cdots + \mathbf{u}_m$$

where each $\mathbf{u}_j \in \mathcal{U}_j$. Then $\mathcal{V}$ is the direct sum of subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_m$, written as $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$

Definition 1.2.6. *Linear Combination*
Let $\mathcal{U}$ be non-empty subset of $\mathbb{F}$-vector space $\mathcal{V}$. A linear combination of elements of $\mathcal{U}$ is expressed as

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_r\mathbf{v}_r$$

where r is a positive integer, and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r \in \mathcal{U}$, $c_1, c_2, \ldots, c_r \in \mathbb{F}$.
A linear combination is *trivial* if $c_1 = c_2 = \cdots = c_r = 0$.

Definition 1.2.7. *Span*
Let $\mathcal{U}$ be subset of $\mathbb{F}$-vector space $\mathcal{V}$. If $\mathcal{U} \neq \emptyset$, then $\text{span}(\mathcal{U})$ is the set of linear combinations of elements of $\mathcal{U}$. Define $\text{span}(\emptyset) = \{\mathbf{0}\}$. If $\beta = \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ is a list of $r \geq 1$ vectors in $\mathcal{V}$, define $\text{span}(\beta) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$.

Lemma 1.2.8. *Linear Dependence Lemma*
If $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly dependent in $\mathcal{V}$ and $\mathbf{v}_1 \neq 0$, then $\exists\ j \in \{2, \ldots, m\}$ such that the following hold:

i. $\mathbf{v}_j \in \mathit{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$

ii. if the jth term is removed from $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$*, the span of the remaining list equals* $\mathit{span}(\mathbf{v}_1, \ldots, \mathbf{v}_m)$.

Proof. i. Suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly dependent in $\mathcal{V}$, and $\mathbf{v}_1 \neq 0$. Then $\exists\ a_1, \ldots, a_m \in \mathbb{F}$, not all 0, s.t.

$$a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = 0$$

Not all of $a_2, \ldots, a_m$ can be 0 as $\mathbf{v}_1 \neq 0$. Let j be largest element of $2, \ldots, m$ s.t. $a_j \neq 0$. Then

$$\mathbf{v}_j = -\frac{a_1}{a_j}\mathbf{v}_1 - \cdots - \frac{a_{j-1}}{a_j}\mathbf{v}_{j-1}$$

ii. Suppose $\boldsymbol{u} \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_m)$. Then $\exists\ c_1, \ldots, c_m \in \mathbb{F}$ such that

$$\mathbf{u} = c_1\mathbf{v}_1 + \cdots + c_m\mathbf{v}_m$$

Replace $\mathbf{v}_j$ with equation from (i), hence (ii) holds.

□

Theorem 1.2.9. *In a finite-dimensional vector space, length of every linearly independent list of vectors is less than or equal to length of every spanning list of vectors.*

Proof. Suppose $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is linearly independent in $\mathcal{V}$, and $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ spans $\mathcal{V}$. Need to prove $m \leq n$.

i. List $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ spans $\mathcal{V}$, hence adjoining any vector to it produces linearly dependent list. By Lemma 1.2.8, can remove one of $\mathbf{w}$ such that the list containing $\mathbf{u}_1$ and remaining $\mathbf{w}$'s spans $\mathcal{V}$.

ii. List of length n from previous step $j-1$ spans $\mathcal{V}$, hence adjoining any vector to it produces a linearly dependent list. The list of length $(n+1)$ obtained by adjoining $\mathbf{u}_j$ to the list, placing it just after $\mathbf{u}_1, \ldots, \mathbf{u}_{j-1}$ is linearly dependent. By Lemma 1.2.8, one of the vectors in this list is span of previous ones, because $(\mathbf{u}_1, \ldots, \mathbf{u}_j)$ is linearly independent, hence this vector must be one of $\mathbf{w}$ not one of $\mathbf{u}$. Hence can remove $\mathbf{w}$ from the list so that the new list consists of $\mathbf{u}_1, \ldots, \mathbf{u}_j$ and remaining $\mathbf{w}$ spans $\mathcal{V}$.

iii. At step m, all $\mathbf{u}$ has been added, and process tops. If at any step $\mathbf{u}$ is added and no more $\mathbf{w}$ could be removed, there is a contradiction. hence there must be at least as many $\mathbf{w}$ as $\mathbf{u}$.

□

Proposition 1.2.10. *Every subspace of finite-dimensional vector space is finite dimensional.*

Proof. Suppose $\mathcal{V}$ is finite dimensional, and $\mathcal{U}$ is subspace of $\mathcal{V}$.

i. If $\mathcal{U} = \{0\}$, then $\mathcal{U}$ is finite dimensional. If $\mathcal{U} \neq \{0\}$, choose nonzero vector $\mathbf{v}_1 \in \mathcal{U}$.

ii. If $\mathcal{U} = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$, then U is finite dimensional. If $\mathcal{U} \neq \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$, choose vector $\mathbf{v}_j \in \mathcal{U}$ such that $\mathbf{v}_j \neq \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$

After each step, a linearly independent list is constructed by Lemma 1.2.8. This linearly independent list cannot be longer than any spanning list of $\mathcal{V}$, hence process must terminate, which means $\mathcal{U}$ is finite dimensional. □

Definition 1.2.11. A *basis* of $\mathcal{V}$ is a list of vectors in $\mathcal{V}$ that is linearly independent and spans $\mathcal{V}$.

Proposition 1.2.12. *A list* $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *of vectors in* $\mathcal{V}$ *is a basis of* $\mathcal{V}$ *if and only if every* $\mathbf{v} \in \mathcal{V}$ *can be written uniquely in form* $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$*, where* $a_1, \ldots, a_n \in \mathbb{F}$.

Proof. First suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is basis of $\mathcal{V}$. Let $\mathbf{v} \in \mathcal{V}$. As $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ spans $\mathcal{V}$, $\exists\ a_1, \ldots, a_n \in \mathbb{F}$ such that proposition holds. To show the representation is unique, suppose $b_1, \ldots, b_n$ are scalars, hence

$$\mathbf{v} = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$$

Subtracting this from proposition equation,

$$0 = (a_1 - b_1)\mathbf{v}_1 + \cdots + (a_n - b_n)\mathbf{v}_n$$

This implies each $a_j - b_j = 0$ as $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent, hence $a_1 = b_1, \ldots, a_n = b_n$.
For other direction, suppose every $\mathbf{v} \in \mathcal{V}$ can be written uniquely in proposition form. To show $(\mathbf{v}_1, \ldots \mathbf{v}_n)$ is linearly independent, suppose $a_1, \ldots a_n \in \mathbb{F}$ are such that

$$0 = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$$

The uniqueness of representation from proposition with $\mathbf{v} = 0$ implies $a_1 = \cdots = a_n = 0$. Hence $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent and hence is a basis of $\mathcal{V}$. □

Theorem 1.2.13. *Every spanning list in a vector space can be reduced to a basis of vector space.*

Proof. Suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ spans $\mathcal{V}$. To remove some vectors from $(\mathbf{v}_1, \ldots \mathbf{v}_n)$ so that remaining vectors form a basis of $\mathcal{V}$. Start with $\mathcal{B} = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$.

i. If $\mathbf{v}_1 = 0$, delete $\mathbf{v}_1$ from B, If $\mathbf{v}_1 \neq 0$, leave $\mathcal{B}$ unchanged.

ii. If $\mathbf{v}_j$ is in span$(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$, delete $\mathbf{v}_j$ from $\mathcal{B}$. If $\mathbf{v}_j$ is not in span$(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$, leave $\mathcal{B}$ unchanged.

Stop process after step n, getting a list $\mathcal{B}$ that spans $\mathcal{V}$, and discarded only vectors that were already in span of previous vectors. Hence $\mathcal{B}$ is linearly independent by Lemma 1.2.8. Hence $\mathcal{B}$ is a basis of $\mathcal{V}$. □

Corollary 1.2.14. *Every finite-dimensional vector space has a basis.*

Proof. By definition, a finite-dimensional vector space has a spanning list. By Theorem 1.2.13, any spanning list can be reduced to a basis. □

Theorem 1.2.15. *Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.*

Proof. Suppose $\mathcal{V}$ is finite dimensional and $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent in $\mathcal{V}$. To extend $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ to a basis of $\mathcal{V}$. Let $(w_1, \ldots, w_n)$ be any list of vectors in $\mathcal{V}$ that spans $\mathcal{V}$.

i. If $\mathbf{w}_1 \in$ span$(\mathbf{v}_1, \ldots, \mathbf{v}_m)$, let $\mathcal{B} = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$. If $\mathbf{w}_1$ is not in span$(\mathbf{v}_1, \ldots, \mathbf{v}_m)$, let $\mathcal{B} = (\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{w}_1)$.

ii. If $\mathbf{w}_j \in$ span(B), leave $\mathcal{B}$ unchanged. If $\mathbf{w}_j$ not in span(B), extend $\mathcal{B}$ by adjoining $\mathbf{w}_j$ to it.

After each step, $\mathcal{B}$ is still linearly independent as otherwise Lemma 1.2.8 would give a contradiction. After step n, the span of $\mathcal{B}$ includes all the $\mathbf{w}$. Hence the $\mathcal{B}$ obtained after step n spans $\mathcal{V}$ and hence is a basis of $\mathcal{V}$. □

This theorem can be applied to show that every subspace of a finite-dimensional vector space can be paired with another subspace to form a direct sum of the whole space.

Proposition 1.2.16. *Suppose $\mathcal{V}$ is finite dimensional and $\mathcal{U}$ is a subspace of $\mathcal{V}$. Then there is a subspace $\mathcal{W}$ of $\mathcal{V}$ such that $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.*

Proof. As $\mathcal{V}$ is finite dimensional, so is $\mathcal{U}$, hence $\exists$ basis$(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$. As $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is a linearly independent list of vectors of $\mathcal{V}$, it can be extended to a basis $(\mathbf{u}_1, \ldots \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_m)$ of $\mathcal{V}$. Let $\mathcal{W} =$ span$(\mathbf{w}_1, \ldots, \mathbf{w}_n)$. To prove $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, need to show that $\mathcal{V} = \mathcal{U} + \mathcal{W}$ and $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$.
Suppose $\mathbf{v} \in \mathcal{V}$. As $\mathbf{u}_1, \ldots, \mathbf{u}_m$ spans $\mathcal{V}$, there exists scalars $a_1, \ldots, a_m, b_1, \ldots, b_n \in \mathbb{F}$ such that

$$\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_m\mathbf{u}_m + b_1\mathbf{w}_1 + \cdots + b_n\mathbf{w}_n = \mathbf{u} + \mathbf{w}$$

where $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{W}$ are defined as above. Hence $\mathbf{v} \in \mathcal{U} + \mathcal{W}$, thus $\mathcal{V} = \mathcal{U} + \mathcal{W}$.
To show $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, suppose $\mathbf{v} \in \mathcal{U} \cap \mathcal{W}$. Then $\exists$ scalars $a_1, \ldots, a_m, b_1, \ldots, b_n \in \mathbb{F}$. such that

$$\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_m\mathbf{u}_m = b_1\mathbf{w}_1 + \cdots + b_n\mathbf{w}_n$$

Hence

$$a_1\mathbf{u}_1 + \cdots + a_m\mathbf{u}_m - b_1\mathbf{w}_1 - \cdots - b_n\mathbf{w}_n = \mathbf{0}$$

As $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ is linearly independent, this implies that $a_1 = \cdots = a_m = b_1 = \cdots = b_n = 0$, hence $\mathbf{v} = \mathbf{0}$, completing proof that $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$. □

Theorem 1.2.17. *Any two bases of a finite-dimensional vector space have the same length.*

Proof. Suppose $\mathcal{V}$ is finite dimensional, and $\mathcal{B}_1, \mathcal{B}_2$ be any two bases of $\mathcal{V}$. Then $\mathcal{B}_1$ is linearly independent in $\mathcal{V}$, and $\mathcal{B}_2$ spans $\mathcal{V}$. Hence length of $\mathcal{B}_1$ is at most the length of $\mathcal{B}_2$. Interchanging roles of $\mathcal{B}_1$ and $\mathcal{B}_2$, the length of $\mathcal{B}_2$ is at most the length of $\mathcal{B}_1$. Hence length of $\mathcal{B}_1$ must equal length of $\mathcal{B}_2$ as desired. □

Definition 1.2.18. The *dimension* of a finite-dimensional vector space is defined to be the length of any basis of the vector space, denoted as dim$(\mathcal{V})$.

Proposition 1.2.19. *If $\mathcal{V}$ is finite dimensional and $\mathcal{U}$ is a subspace of $\mathcal{V}$, then dim$(\mathcal{U}) \leq$ dim$(\mathcal{V})$.*

Proof. Suppose $\mathcal{V}$ is finite dimensional and $\mathcal{U}$ is a subspace of $\mathcal{V}$. Any basis of $\mathcal{U}$ is a linearly independent list of vectors in $\mathcal{V}$, thus can be extended to a basis of $\mathcal{V}$. Hence length of a basis of $\mathcal{U} \leq$ length of a basis of $\mathcal{V}$. □

Proposition 1.2.20. *If $\mathcal{V}$ is finite dimensional, then every spanning list of vectors in $\mathcal{V}$ with length $dim(\mathcal{V})$ is a basis of $\mathcal{V}$.*

Proof. Suppose $\dim(\mathcal{V}) = n$ and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ spans $\mathcal{V}$. The list $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ can be reduced to a basis of $\mathcal{V}$. However, every basis of $\mathcal{V}$ has length n, hence reduction must be trivial, i.e. no elements are deleted from $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$. Hence, $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ as desired. □

Proposition 1.2.21. *If $\mathcal{V}$ is finite dimensional, then every linearly independent list of vectors in $\mathcal{V}$ with length $dim(\mathcal{V})$ is a basis of $\mathcal{V}$.*

Proof. Suppose $\dim(\mathcal{V}) = n$ and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent in $\mathcal{V}$. The list $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ can be extended to a basis of $\mathcal{V}$. However, every basis of $\mathcal{V}$ has length n, hence the extension must be trivial, i.e., no new elements are adjoined to $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$. Hence, $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ as desired. □

The next theorem gives formula for dimension of sum of two subspaces of a finite-dimensional vector space.

Theorem 1.2.22. *If $\mathcal{U}_1$ and $\mathcal{U}_2$ are subspaces of a finite-dimensional vector space, then*

$$dim(\mathcal{U}_1 + \mathcal{U}_2) = dim(\mathcal{U}_1) + dim(\mathcal{U}_2) - dim(\mathcal{U}_1 \cap \mathcal{U}_2)$$

Proof. Let $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ be basis of $\mathcal{U}_1 \cap \mathcal{U}_2$, hence $\dim(\mathcal{U}_1 \cap \mathcal{U}_2) = m$.
As $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is a basis of $\mathcal{U}_1 \cap \mathcal{U}_2$, it is linearly independent in $\mathcal{U}_1$, hence can be extended to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_j)$ of $\mathcal{U}_1$. Hence $\dim(\mathcal{U}_1) = m+j$. Extend $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ to a basis of $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_k)$ of $\mathcal{U}_2$, hence $\dim(\mathcal{U}_2) = m + k$.
Note that $\text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_j, \mathbf{w}_1, \ldots, \mathbf{w}_k)$ contains $\mathcal{U}_1$ and $\mathcal{U}_2$, hence contains $\mathcal{U}_1 + \mathcal{U}_2$. To show that this list is a basis of $\mathcal{U}_1 + \mathcal{U}_2$, need to show that it is linearly independent. Suppose

$$a_1\mathbf{u}_1 + \cdots + a_m\mathbf{u}_m + b_1\mathbf{v}_1 + \cdots + b_j\mathbf{v}_j + c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k = \mathbf{0}$$

where $a_i, b_i, c_i \in \mathbb{F}$. Rewriting the equation,

$$c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k = -a_1\mathbf{u}_1 - \cdots - a_m\mathbf{u}_m - b_1\mathbf{v}_1 - \cdots - b_j\mathbf{v}_j$$

which shows $c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k \in \mathcal{U}_1$. All $\mathbf{w}$ are in $\mathcal{U}_2$, hence this implies $c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k \in \mathcal{U}_1 \cap \mathcal{U}_2$. As $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is a basis of $\mathcal{U}_1 \cap \mathcal{U}_2$, write this as

$$c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k = d_1\mathbf{u}_1 + \cdots + d_m\mathbf{u}_m$$

for some $d_i \in \mathbb{F}$. As $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_k)$ is linearly independent, the last equation implies all c_i and d_i equal 0. Hence the original equation becomes

$$a_1\mathbf{u}_1 + \cdots + a_m\mathbf{u}_m + b_1\mathbf{v}_1 + \cdots + b_j\mathbf{v}_j = \mathbf{0}$$

This implies all a_i and b_i are 0 as the list $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_k)$ is linearly independent.
Hence $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_j, \mathbf{w}_1, \ldots, \mathbf{w}_k)$ is a basis of $\mathcal{U}_1 + \mathcal{U}_2$, and thus

$$\begin{aligned}\dim(\mathcal{U}_1 + \mathcal{U}_2) &= m + j + k \\ &= (m + j) + (m + k) - m \\ &= \dim(\mathcal{U}_1) + \dim(\mathcal{U}_2) - \dim(\mathcal{U}_1 \cap \mathcal{U}_2)\end{aligned}$$

□

Note that dimension meshes well with direct sums, which is analogous to disjoint union.

Proposition 1.2.23. *Suppose $\mathcal{V}$ is finite dimensional and $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$ such that*

$$\begin{aligned}\mathcal{V} &= \mathcal{U}_1 + \cdots + \mathcal{U}_m \\ dim(\mathcal{V}) &= dim(\mathcal{U}_1) + \cdots + dim(\mathcal{U}_m)\end{aligned}$$

Then $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Proof. Choose a basis for each $\mathcal{U}_j$. Put these bases together in a list which spans $\mathcal{V}$ and has length $\dim(\mathcal{V})$. Hence this list is a basis of $\mathcal{V}$, and it is linearly independent. Suppose $\mathbf{u}_1 \in \mathcal{U}_1, \ldots, \mathbf{u}_m \in \mathcal{U}_m$ such that

$$\mathbf{0} = \mathbf{u}_1 + \cdots + \mathbf{u}_m$$

Write each $\mathbf{u}_j$ as a linear combination of basis vectors of $\mathcal{U}_j$. Substitute these linear combinations into equation above, hence $\mathbf{0}$ is written as a linear combination of basis of $\mathcal{V}$ constructed above. Hence all scalars used in this linear combination must be 0. Thus each $\mathbf{u}_j = \mathbf{0}$, which proves $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$. □

1.2.2 Linear Maps

Definition 1.2.24. A *Linear Map* from $\mathcal{V}$ to $\mathcal{W}$ is a function $T : \mathcal{V} \to \mathcal{W}$ with following properties:

i. *Additivity*: $T(\mathbf{u} + \mathbf{v}) = T\mathbf{u} + T\mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$

ii. *Homogeneity*: $T(a\mathbf{v}) = a(T\mathbf{v})$ for all $a \in \mathbb{F}$ and all $\mathbf{v} \in \mathcal{V}$.

The set of all linear maps from $\mathcal{V}$ to $\mathcal{W}$ is denoted $\mathcal{L}(\mathcal{V}, \mathcal{W})$.
The linear map is *injective* if whenever $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $T\mathbf{u} = T\mathbf{v}$, that $\mathbf{u} = \mathbf{v}$.
The linear map is *surjective* if its range equals $\mathcal{W}$.

Example 1.2.25. Examples of linear maps include

i. Zero Map: $0 \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is defined by $0\mathbf{v} = 0$.

ii. Identity Map: $I \in \mathcal{L}(\mathcal{V}, \mathcal{V})$ is defined by $I\mathbf{v} = \mathbf{v}$

iii. Differentiation: $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}), \mathcal{P}(\mathbf{R}))$ is defined by $T\mathbf{p} = \mathbf{p}'$

iv. Integration: $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}), \mathbf{R})$ is defined by $T\mathbf{p} = \int_0^1 \mathbf{p}(\mathbf{x})\, d\mathbf{x}$

v. Backward Shift: Let $\mathbb{F}^\infty$ be vector space of all sequences of elements of $\mathbb{F}$. Let $T \in \mathcal{L}(\mathbb{F}^\infty, \mathbb{F}^\infty)$. The backward shift map is then $T(x_1, x_2, x_3, \ldots) = (x_2, x_3, \ldots)$

vi. From $\mathbb{F}^n$ to $\mathbb{F}^m$: $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ defined by $T(x_1, \ldots, x_n) = (a_{1,1}x_1 + \cdots + a_{1,n}x_n, \ldots, a_{m,1}x_1 + \cdots + a_{m,n}x_n)$, where $a_{j,k} \in \mathbb{F}$, $j = 1, \ldots, m$ and $k = 1, \ldots, n$.

Remark 1.2.26. *Properties of Product of Linear Maps*
Let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. The *product* of S and T is ST.

i. Associativity: $(T_1T_2)T_3 = T_1(T_2T_3)$

ii. Identity: $TI = T$ (identity map on $\mathcal{V}$) and $IT = T$ (identity map on $\mathcal{W}$)

iii. Distributive: $(S_1 + S_2)T = S_1T + S_2T$ and $S(T_1 + T_2) = ST_1 + ST_2$ whenever $T, T_1, T_2 \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S, S_1, S_2 \in \mathcal{L}(\mathcal{V}, \mathcal{W})$

Multiplication of linear maps is not commutative.

Definition 1.2.27. For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *null space* of T, denoted $\text{null}(T)$, is the subset of $\mathcal{V}$ consisting of those vectors that maps T to 0, $\text{null}(T) = \{\mathbf{v} \in \mathcal{V} : T\mathbf{v} = \mathbf{0}\}$.

Proposition 1.2.28. *If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then null$(T)$ is a subspace of $\mathcal{V}$.*

Proof. Suppose $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By additivity,

$$T(\mathbf{0}) = T(\mathbf{0} + \mathbf{0}) = T(\mathbf{0}) + T(\mathbf{0})$$

which implies $T(\mathbf{0}) = \mathbf{0}$. Hence $\mathbf{0} \in \text{null}(T)$.
If $\mathbf{u}, \mathbf{v} \in \text{null}(T)$, then

$$T(\mathbf{u} + \mathbf{v}) = T\mathbf{u} + T\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

Hence $\mathbf{u} + \mathbf{v} \in \text{null}(T)$, and $\text{null}(T)$ is closed under addition.
If $\mathbf{u} \in \text{null}(T)$ and $a \in \mathbb{F}$, then

$$T(a\mathbf{u}) = aT\mathbf{u} = a\mathbf{0} = \mathbf{0}$$

Hence $a\mathbf{u} \in \text{null}(T)$, and $\text{null}(T)$ is closed under scalar multiplication.
As $\text{null}(T)$ contains $\mathbf{0}$ and is closed under addition and scalar multiplication, $\text{null}(T)$ is subspace of $\mathcal{V}$. □

Proposition 1.2.29. *Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is injective if and only if null$(T) = \{\mathbf{0}\}$.*

Proof. ($\Rightarrow$) Suppose T is injective. By Proposition 1.2.28, $\{\mathbf{0}\} \subset \text{null}(T)$. Suppose $\mathbf{v} \in \text{null}(T)$. Then $T(\mathbf{v}) = \mathbf{0} = T(\mathbf{0})$. As T is injective, equation above implies $\mathbf{v} = \mathbf{0}$. Thus $\text{null}(T) = \{\mathbf{0}\}$.
($\Leftarrow$) Suppose $\text{null}(T) = \{\mathbf{0}\}$, $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $T\mathbf{u} = T\mathbf{v}$. Then $\mathbf{0} = T\mathbf{u} - T\mathbf{v} = T(\mathbf{u} - \mathbf{v})$. Hence $\mathbf{u} - \mathbf{v}$ is in $\text{null}(T)$, which equals $\{\mathbf{0}\}$. Thus $\mathbf{u} - \mathbf{v} = \mathbf{0}$, which implies $\mathbf{u} = \mathbf{v}$. Hence T is injective. □

Definition 1.2.30. For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *range* of T, denoted $\text{range}(T)$, is subset of $\mathcal{W}$ consisting of vectors that are of from $T\mathbf{v}$ for some $\mathbf{v} \in \mathcal{V}$, where $\text{range}(T) = \{T\mathbf{v} : \mathbf{v} \in \mathcal{V}\}$.

Proposition 1.2.31. *If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then range$(T)$ is a subspace of $\mathcal{W}$.*

Proof. Suppose $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T(\mathbf{0}) = \mathbf{0}$ by Proposition 1.2.28, implying $\mathbf{0} \in \text{range}(T)$.
If $\mathbf{w}_1, \mathbf{w}_2 \in \text{range}(T)$, then $\exists\ \mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$ such that $T\mathbf{v}_1 = \mathbf{w}_1$ and $T\mathbf{v}_2 = \mathbf{w}_2$. Hence

$$T(\mathbf{v}_1 + \mathbf{v}_2) = T\mathbf{v}_1 + T\mathbf{v}_2 = \mathbf{w}_1 + \mathbf{w}_2$$

thus $\mathbf{w}_1 + \mathbf{w}_2 \in \text{range}(T)$. Hence $\text{range}(T)$ is closed under addition.
If $\mathbf{w} \in \text{range}(T)$ and $a \in \mathbb{F}$, then $\exists\ \mathbf{v} \in \mathcal{V}$ such that $T\mathbf{v} = \mathbf{w}$. Hence $T(a\mathbf{v}) = aT\mathbf{v} = a\mathbf{w}$, and $a\mathbf{w} \in \text{range}(T)$. Thus $\text{range}(T)$ is closed under scalar multiplication.
As $\text{range}(T)$ contains $\mathbf{0}$ and is closed under addition and scalar multiplication, $\text{range}(T)$ is a subspace of $\mathcal{W}$. □

The next theorem states that the dimension of the null space plus dimension of range of linear map on finite-dimensional vector space equals dimension of the domain.

Theorem 1.2.32. *If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then range$(T)$ is a finite dimensional subspace of $\mathcal{W}$, and dim$(\mathcal{V})$ = dim(null(T)) + dim(range(T)).*

Proof. Suppose $\mathcal{V}$ is a finite-dimensional vector space and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Let $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ be a basis of $\text{null}(T)$, thus $\dim(\text{null}(T)) = m$. The linearly independent list $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ can be extended to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$. Hence $\dim(\mathcal{V}) = m + n$. Need to show $\text{range}(T)$ is finite dimensional and $\dim(\text{range}(T)) = n$, by proving that $(T\mathbf{w}_1, \ldots, T\mathbf{w}_n)$ is a basis of $\text{range}(T)$.
Let $\mathbf{v} \in \mathcal{V}$. As $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ spans $\mathcal{V}$, then

$$\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_m\mathbf{u}_m + b_1\mathbf{w}_1 + \cdots + b_n\mathbf{w}_n$$

where $a, b \in \mathbb{F}$. Applying T to both sides of equation,

$$T\mathbf{v} = b_1T\mathbf{w}_1 + \cdots + b_nT\mathbf{w}_n$$

Note $T\mathbf{u}_j$ disappeared as each $\mathbf{u}_j \in \text{null}(T)$. Last equation implies $(T\mathbf{w}_1, \ldots T\mathbf{w}_n)$ spans $\text{range}(T)$, hence that $\text{range}(T)$ is finite dimensional.
To show $(T\mathbf{w}_1, \ldots T\mathbf{w}_n)$ is linearly independent, suppose $c_1, \ldots, c_n \in \mathbb{F}$, and

$$\begin{aligned} c_1T\mathbf{w}_1 + \cdots + c_nT\mathbf{w}_n &= \mathbf{0} \\ T(c_1\mathbf{w}_1 + \cdots + c_n\mathbf{w}_n) &= \mathbf{0} \\ c_1\mathbf{w}_1 + \cdots + c_n\mathbf{w}_n &\in \text{null}(T) \end{aligned}$$

As $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ spans $\text{null}(T)$, then

$$c_1\mathbf{w}_1 + \cdots + c_n\mathbf{w}_n = d_1\mathbf{u}_1 + \cdots + d_m\mathbf{u}_m$$

where $d_i \in \mathbb{F}$. This implies all c and d are 0, as $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ are linearly independent.
Hence $(T\mathbf{w}_1, \ldots, T\mathbf{w}_n)$ is linearly independent and is a basis for $\text{range}(T)$. □

No linear map from finite-dimensional vector space to a smaller vector space can be injective.

Corollary 1.2.33. *If $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces such that dim$(\mathcal{V})$ > dim$(\mathcal{W})$, then no linear map from $\mathcal{V}$ to $\mathcal{W}$ is injective.*

Proof. Suppose $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces such that $\dim(\mathcal{V}) > \dim(\mathcal{W})$. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$,

$$\begin{aligned} \dim(\text{null}(T)) &= \dim(\mathcal{V}) - \dim(\text{range}(T)) \\ &\geq \dim(\mathcal{V}) - \dim(\mathcal{W}) \geq 0 \end{aligned}$$

As $\dim(\text{null}(T)) \geq 0$, $\text{null}(T)$ must contain vectors other than 0, hence T is not injective. □

No linear map from finite-dimensional vector space to a bigger vector space can be surjective.

Corollary 1.2.34. *If $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces such that $\dim(\mathcal{V}) < \dim(\mathcal{W})$, then no linear map from $\mathcal{V}$ to $\mathcal{W}$ is surjective.*

Proof. Suppose $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces such that $\dim(\mathcal{V}) < \dim(\mathcal{W})$. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$,

$$\begin{aligned}\dim(\text{range}(T)) &= \dim(\mathcal{V}) - \dim(\text{null}(T)) \\ &\leq \dim(\mathcal{V}) < \dim(\mathcal{W})\end{aligned}$$

As $\dim(\text{range}(T)) < \dim(\mathcal{W})$, $\text{range}(T)$ cannot equal $\mathcal{W}$, hence T is not surjective. □

Definition 1.2.35. A $m \times n$ *matrix* $\mathcal{M}$ has m rows and n columns,

$$\begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix}$$

Proposition 1.2.36. *Suppose $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$, and $(\mathbf{w}_1, \ldots, \mathbf{w}_m)$ is a basis of $\mathcal{W}$. Then $\mathcal{M}(T\mathbf{v}) = \mathcal{M}(T)\mathcal{M}(\mathbf{v})$ for every $\mathbf{v} \in \mathcal{V}$.*

Proof. Let

$$\begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix}$$

Note that

$$T\mathbf{v}_k = \sum_{j=1}^{m} a_{j,k}\mathbf{w}_j$$

for each k. Let $\mathbf{v}$ be an arbitrary vector in $\mathcal{V}$, then $\mathcal{M}(\mathbf{v})$ is given by

$$\begin{aligned} T\mathbf{v} &= b_1 T\mathbf{v}_1 + \cdots + b_n T\mathbf{v}_n \\ &= b_1 \sum_{j=1}^{m} a_{j,1}\mathbf{w}_j + \cdots + b_n \sum_{j=1}^{m} a_{j,n}\mathbf{w}_j \\ &= \sum_{j=1} (a_{j,1}b_1 + \cdots + a_{j,n}b_n)\mathbf{w}_j \end{aligned}$$

The equation shows that $\mathcal{M}(T\mathbf{v})$ is a $m \times 1$ matrix of vector $T\mathbf{v}$ with respect to basis $(\mathbf{w}_1, \ldots, \mathbf{w}_m)$, given by

$$\mathcal{M}(T\mathbf{v}) = \begin{bmatrix} a_{1,1}b_1 + \cdots + a_{1,n}b_n \\ \vdots \\ a_{m,1}b_1 + \cdots + a_{m,n}b_n \end{bmatrix}$$

Hence $\mathcal{M}(T\mathbf{v}) = \mathcal{M}(T)\mathcal{M}(\mathbf{v})$. □

Definition 1.2.37. A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is *invertible* if $\exists$ linear map $S \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ such that ST equals identity map on $\mathcal{V}$ and TS equals identity map on $\mathcal{W}$.
A linear map $S \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ satisfying $ST = I$ and $TS = I$ is the *inverse* of T.

Proposition 1.2.38. *A linear map is invertible if and only if it is injective and surjective.*

Proof. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
($\Rightarrow$) Suppose T is invertible, $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, and $T\mathbf{u} = T\mathbf{v}$. Then

$$\mathbf{u} = T^{-1}(T\mathbf{u}) = T^{-1}T(\mathbf{v}) = \mathbf{v}$$

hence $\mathbf{u} = \mathbf{v}$. Thus T is injective.
To show T is surjective, let $\mathbf{w} \in \mathcal{W}$, then $\mathbf{w} = T(T^{-1}\mathbf{w})$, thus $\mathbf{w}$ is in range of T. Hence $\text{range}(T) = W$.
Suppose T is injective and surjective, to prove T is invertible. For each $\mathbf{w} \in \mathcal{W}$, define $S\mathbf{w}$ to be unique element

of $\mathcal{V}$ such that $T(S\mathbf{w}) = \mathbf{w}$, which shows TS equals identity map on $\mathcal{W}$. To prove ST equals identity map on $\mathcal{V}$, let $\mathbf{v} \in \mathcal{V}$, then

$$T(ST\mathbf{v}) = (TS)(T\mathbf{v}) = I(T\mathbf{v}) = T\mathbf{v}$$

which implies $ST\mathbf{v} = \mathbf{v}$ (as T is injective), thus ST equals identity map on $\mathcal{V}$. Let $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$, then

$$T(S\mathbf{w}_1 + S\mathbf{w}_2) = T(S\mathbf{w}_1) + T(S\mathbf{w}_2) = \mathbf{w}_1 + \mathbf{w}_2$$

Thus $S\mathbf{w}_1 + S\mathbf{w}_2$ is unique element of $\mathcal{V}$ that maps T to $\mathbf{w}_1 + \mathbf{w}_2$. This implies S satisfies the additive property requires for linearity.
For proof of homogeneity, let $\mathbf{w} \in \mathcal{W}$ and $a \in \mathbb{F}$, then

$$T(aS\mathbf{w}) = aT(S\mathbf{w}) = a\mathbf{w}$$

Thus $aS\mathbf{w}$ is the unique element of $\mathcal{V}$ that T maps to $a\mathbf{w}$. By definition of S, this implies $S(a\mathbf{w}) = aS\mathbf{w}$, hence S is linear, as desired. □

Definition 1.2.39. Two vector spaces are *isomorphic* if there is an invertible linear map from one vector space onto the other. Two isomorphic spaces have the same properties.

Theorem 1.2.40. *Two finite-dimensional vector spaces are isomorphic if and only if they have the same dim.*

Proof. ($\Rightarrow$) Suppose $\mathcal{V}$ and $\mathcal{W}$ are isomorphic finite-dimensional vector spaces.
$\exists$ invertible linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. As T is invertible, $\text{null}(T) = \{\mathbf{0}\}$, and $\text{range}(T) = \mathcal{W}$, thus $\dim(\text{null}(T)) = 0$ and $\dim(\text{range}(T)) = \dim(\mathcal{W})$.

$$\begin{aligned}\dim(\mathcal{V}) &= \dim(\text{null}(T)) + \dim(\text{range}(T)) \\ &= \dim(\mathcal{W})\end{aligned}$$

($\Leftarrow$) Suppose $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces with the same dimension. let $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be basis of $\mathcal{V}$ and $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be basis of $\mathcal{W}$. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ defined by

$$T(a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n) = a_1\mathbf{w}_1 + \cdots + a_n\mathbf{w}_n$$

Then T is surjective as $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ spans $\mathcal{W}$, and T is injective as $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ is linearly independent.
As T is injective and surjective, it is invertible, hence $\mathcal{V}$ and $\mathcal{W}$ are isomorphic. □

The next theorem implies that every finite-dimensional vector space is isomorphic to some $\mathbb{F}^n$.

Theorem 1.2.41. *Suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ and $(\mathbf{w}_1, \cdots, \mathbf{w}_m)$ is a basis of $\mathcal{W}$. Then $\mathcal{M}$ is an invertible linear map between $\mathcal{L}(\mathcal{V}, \mathcal{W})$ and Mat$(m, n, \mathbb{F})$,*

Proof. Note that $\mathcal{M}$ is linear, hence to prove that $\mathcal{M}$ is injective and surjective.
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $\mathcal{M}(T) = 0$, then $T\mathbf{v}_k = \mathbf{0}$ for $k = 1, \ldots, n$, As $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$, this implies $T = 0$, hence $\mathcal{M}$ is injective.
To show $\mathcal{M}$ is surjective, let

$$A = \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix}$$

be a matrix in $\text{Mat}(m, n, \mathbb{F})$. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ such that

$$T\mathbf{v}_k = \sum_{j=1}^{m} a_{j,k}\mathbf{w}_k$$

for $k = 1, \ldots, n$. Note $\mathcal{M}(T) = A$, so range of $\mathcal{M}$ equals $\text{Mat}(m, n, \mathbb{F})$ as desired. □

Proposition 1.2.42. *If $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional, then $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is finite-dimensional, and*

$$dim\mathcal{L}(\mathcal{V}, \mathcal{W}) = (dim(\mathcal{V}))(dim(\mathcal{W}))$$

Proof. This follows from $\dim(\text{Mat}(m, n, \mathbb{F})) = mn$ and Theorem 1.2.40 and Theorem 1.2.41. □

Definition 1.2.43. A linear map from a vector space to itself is an *operator*, denoted $\mathcal{L}(\mathcal{V})$.

Theorem 1.2.44. *Suppose $\mathcal{V}$ is finite-dimensional. If $T \in \mathcal{L}(\mathcal{V})$, then the following are equivalent:*

i. T is invertible.

ii. T is injective.

iii. T is surjective.

Proof. Suppose $T \in \mathcal{L}(\mathcal{V})$, then $(i) \Rightarrow (ii)$.
Suppose (ii) holds so that T is injective. Thus $\text{null}(T) = \{\mathbf{0}\}$. Then

$$\begin{aligned}\dim(\text{range}(T)) &= \dim(\mathcal{V}) - \dim(\text{null}(T)) \\ &= \dim(\mathcal{V})\end{aligned}$$

which implies range of T equals $\mathcal{V}$, hence T is surjective, and $(ii) \Rightarrow (iii)$.
Suppose (iii) holds, so T is surjective. Thus $\text{range}(T) = \mathcal{V}$. Then

$$\dim(\text{null}(T)) = \dim(\mathcal{V}) - \dim(\text{range}(T)) = 0$$

which implies $\text{null}(T)$ equals $\{\mathbf{0}\}$, hence T is injective and thus invertible. Hence $(iii) \Rightarrow (i)$. □

1.2.3 Eigenvalues and Eigenvectors

Definition 1.2.45. Let $T \in \mathcal{L}(\mathcal{V})$, and $\exists$ direct sum decomposition $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$ where each $\mathcal{U}_j$ is proper subspace of $\mathcal{V}$. The *restriction* of T to smaller domain $\mathcal{U}_j$ is denoted $T|_{\mathcal{U}_j}$

Definition 1.2.46. For $T \in \mathcal{L}(\mathcal{V})$ and $\mathcal{U}$ a subspace of $\mathcal{V}$, $\mathcal{U}$ is invariant under T if $\mathbf{u} \in \mathcal{U}$ implies $T\mathbf{u} \in \mathcal{U}$ ($\mathcal{U}$ is invariant under T if $T|_{\mathcal{U}}$ is an operator under $\mathcal{U}$.

Definition 1.2.47. A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of $T \in \mathcal{L}(\mathcal{V})$ if $\exists$ a nonzero vector $\mathbf{u} \in \mathcal{V}$ such that $T\mathbf{u} = \lambda\mathbf{u}$. A vector $\mathbf{u} \in \mathcal{V}$ is an *eigenvector* of T if $T\mathbf{u} = \lambda\mathbf{u}$.

Note that the set of eigenvectors of T corresponding to λ equals $\text{null}(T - \lambda I)$.

Theorem 1.2.48. *Let $T \in \mathcal{L}(\mathcal{V})$, and $\lambda_1, \ldots, \lambda_m$ distinct eigenvalues of T, $\mathbf{v}_1, \ldots, \mathbf{v}_m$ are corresponding nonzero eigenvectors, then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.*

Proof. Suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly dependent. Let k be smallest positive integer such that

$$\mathbf{v}_k \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$$

The existence of k with this property follows from Lemma 1.2.8. Hence $\exists\ a_1, \ldots, a_{k-1} \in \mathbb{F}$ such that

$$\mathbf{v}_k = a_1\mathbf{v}_1 + \cdots + a_{k-1}\mathbf{v}_{k-1}$$

Applying T to both sides of equation gives

$$\lambda_k\mathbf{v}_k = a_1\lambda_1\mathbf{v}_1 + \cdots + a_{k-1}\lambda_{k-1}\mathbf{v}_{k-1}$$

Multiply both sides of second equation by λ_k then subtracting above gives

$$0 = a_1(\lambda_k - \lambda_1)\mathbf{v}_1 + \cdots + a_{k-1}(\lambda_k - \lambda_{k-1})\mathbf{v}_{k-1}$$

As k is chosen to be smallest positive integer satisfying first equation, $(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ is linearly dependent, hence equation above implies that all a are 0, which also means $\mathbf{v}_k = \mathbf{0}$, contracting hypothesis that all $\mathbf{v}$ are nonzero. Hence assumption that $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly dependent must be false. □

Corollary 1.2.49. *Each operator on $\mathcal{V}$ has at most $\dim(\mathcal{V})$ distinct eigenvalues.*

Proof. Let $T \in \mathcal{L}(\mathcal{V})$, and $\lambda_1, \ldots, \lambda_m$ distinct eigenvalues of T. Let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding nonzero eigenvectors. Theorem 1.2.48 implies $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent, hence $m \leq \dim(\mathcal{V})$. □

Theorem 1.2.50. *Every operator on a finite-dimensional, nonzero complex vector space has an eigenvalue.*

Proof. Suppose $\mathcal{V}$ is a complex vector space with $\dim(\mathcal{V}) = n > 0$ and $T \in \mathcal{L}(\mathcal{V})$. Choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$, then $(\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \ldots, T^n\mathbf{v})$ cannot be linearly independent as $\mathcal{V}$ has dimension n and $n+1$ vectors. Thus $\exists\ a_0, \ldots, a_n \in \mathbb{C}$ not all 0 such that

$$0 = a_0\mathbf{v} + a_1T\mathbf{v} + \cdots + a_nT^n\mathbf{v}$$

Let m be largest index such that $a_m \neq 0$. As $\mathbf{v} \neq \mathbf{0}$, the coefficients $a_1, \ldots, a_m$ cannot all be 0, hence $0 < m \leq n$. Let a e coefficients of a polynomial in factored form

$$a_0 + a_1z + \cdots + a_nz^n = c(z - \lambda_1)\cdots(z - \lambda_m)$$

where c is nonzero complex number, each $\lambda_j \in \mathbb{C}$, and equation holds for all $z \in \mathbb{C}$, then

$$\begin{aligned}\mathbf{0} &= a_0\mathbf{v} + a_1T\mathbf{v} + \cdots + a_nT^n\mathbf{v} \\ &= (a_0I + a_1T + \cdots + a_nT^n)\mathbf{v} \\ &= c(T - \lambda_1I)\cdots(T - \lambda_mI)\mathbf{v}\end{aligned}$$

which means $T - \lambda_jI$ is not injective for at least one j. Hence T has an eigenvalue. □

Definition 1.2.51. *Upper triangular matrices* are of the form

$$\begin{bmatrix}\lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n\end{bmatrix}$$

Next proposition shows useful connection between upper-triangular matrices and invariant subspaces.

Proposition 1.2.52. *Suppose $T \in \mathcal{L}(\mathcal{V})$, and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$. Following are equivalent:*

i. The matrix of T with respect to $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is upper triangular.

ii. $T\mathbf{v}_k \in$ span$(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ for each $k = 1, \ldots, n$

iii. span$(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is invariant under T for each $k = 1, \ldots, n$

Proof. $(i) \Rightarrow (ii)$ follows from definitions. Also, $(iii) \Rightarrow (ii)$ as (iii) implies (ii).
$(ii) \Rightarrow (iii)$, suppose (ii) holds. Fix $k \in \{1, \ldots, n\}$. From (ii), we know that

$$\begin{aligned}T\mathbf{v}_1 &\in \text{span}(\mathbf{v}_1) \subset \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k) \\ T\mathbf{v}_2 &\in \text{span}(\mathbf{v}_1, \mathbf{v}_2) \subset \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k) \\ &\vdots \\ T\mathbf{v}_k &\in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)\end{aligned}$$

Hence if $\mathbf{v}$ is a linear combination of $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$, then

$$T\mathbf{v} \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$$

Hence, span$(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ is invariant under T. □

Next theorem shows that for each operator on a complex vector space, there is a basis of the vector space with respect to which the matrix of operator only has 0 below the diagonal.

Theorem 1.2.53. *Suppose $\mathcal{V}$ is a complex vector space and $T \in \mathcal{L}(\mathcal{V})$, then T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$.*

Proof. Note that result holds if $\dim(\mathcal{V}) = 1$.
Suppose $\dim(\mathcal{V}) > 1$ and the result holds for all complex vector spaces with dimension less than $\dim(\mathcal{V})$. Let λ be any eigenvalue of T, and let $\mathcal{U} = \text{range}(T - \lambda I)$. As $T - \lambda I$ is not surjective, $\dim(\mathcal{U}) < \dim(\mathcal{V})$, hence $\mathcal{U}$ is invariant under T. To show this, let $\mathbf{u} \in \mathcal{U}$, then

$$T\mathbf{u} = (T - \lambda I)\mathbf{u} + \lambda\mathbf{u}$$

$(T - \lambda I)\mathbf{u} \in \mathcal{U}$ by definition, and $\lambda\mathbf{u} \in \mathcal{U}$. Hence, equation shows $T\mathbf{u} \in \mathcal{U}$, and $\mathcal{U}$ is invariant under T.
Hence $T|_{\mathcal{U}}$ is an operator on $\mathcal{U}$. By induction, $\exists$ basis $(\mathbf{u}_1, \ldots \mathbf{u}_m)$ of $\mathcal{U}$ with respect to which $T|_{\mathcal{U}}$ has an

upper-triangular matrix. Hence for each j,

$$T\mathbf{u}_j = (T|_{\mathcal{U}})(\mathbf{u}_j) \in \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_j)$$

Extend $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ to basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_n)$ of $\mathcal{V}$. For each k,

$$T\mathbf{v}_k = (T - \lambda I)\mathbf{v}_k + \lambda \mathbf{v}_k$$

Definition of $\mathcal{U}$ shows that $(T - \lambda I)\mathbf{v}_k \in \mathcal{U} = \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_m)$. Hence,

$$T\mathbf{v}_k \in \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_k)$$

Hence T has an upper triangular matrix with respect to basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_k)$. □

Proposition 1.2.54. *Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with respect to some basis of $\mathcal{V}$. Then T is invertible if and only if all entries of the diagonal of upper-triangular matrix are non-zero.*

Proof. Let $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be basis of $\mathcal{V}$ with respect to T which has upper-triangular matrix

$$\mathcal{M}(T, (\mathbf{v}_1, \ldots, \mathbf{v}_n)) = \begin{bmatrix} \lambda_1 & & * \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

($\Rightarrow$) If $\lambda_1 = 0$, then $T\mathbf{v}_1 = \mathbf{0}$, hence T is not invertible
Suppose $1 < k \leq n$ and $\lambda_k = 0$, then T maps each of the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$ into $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$. As $\lambda_k = 0$, the matrix representation implies $T\mathbf{v}_k \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$. Hence, define a linear map

$$S : \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k) \to \text{span}(\mathbf{v}_1, \mathbf{v}_{k-1})$$

by $S\mathbf{v} = T\mathbf{v}$ for $\mathbf{v} \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$. Then S is T restricted to $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$.
Note $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ has dimension k and $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ has dimension $k-1$ (as $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent). As $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ has larger dimension than $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$, no linear map from $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ to $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ is injective, hence there exists nonzero vector $\mathbf{v} \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ such that $S\mathbf{v} = \mathbf{0}$. Hence $T\mathbf{v} = \mathbf{0}$, and hence T is not invertible.
($\Leftarrow$) Suppose T is not invertible. Thus T is not injective, and hence there exists a nonzero vector $\mathbf{v} \in \mathcal{V}$ such that $T\mathbf{v} = \mathbf{0}$. As $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$, then $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k$ where $a_1, \ldots, a_k \in \mathbb{F}$ and $a_k \neq 0$ (represent $\mathbf{v}$ as linear combination of $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ then choose k to be largest index with nonzero coefficient).

$$\begin{aligned} \mathbf{0} = T\mathbf{v} &= T(a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k) \\ &= (a_1T\mathbf{v}_1 + \cdots + a_{k-1}T\mathbf{v}_{k-1}) + a_kT\mathbf{v}_k \end{aligned}$$

The last term in parentheses is in $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ (due to upper-triangular form).
Hence this shows $a_kT\mathbf{v}_k \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$. Multiply by $1/a_k$, which is allowed as $a_k \neq 0$, conclude that $T\mathbf{v}_k \in \text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$. Thus when $T\mathbf{v}_k$ is written as linear combination of the basis $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$, the coefficient of $\mathbf{v}_k$ will be 0, hence λ_k must be 0, completing the proof. □

Proposition 1.2.55. *Suppose $T \in \mathcal{L}(\mathcal{V})$ has upper-triangular matrix with respect to some basis of $\mathcal{V}$, then eigenvalues of T consists of precisely the entries of the diagonal of the upper-triangular matrix.*

Proof. Suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ with respect to T which T has an upper-triangular matrix

$$\mathcal{M}(T, (\mathbf{v}_1, \ldots, \mathbf{v}_n)) = \begin{bmatrix} \lambda_1 & & * \\ & \lambda_2 & \\ & & \ddots \\ 0 & & & \lambda_n \end{bmatrix}$$

Let $\lambda \in \mathbb{F}$, then

$$\mathcal{M}(T - \lambda I, (\mathbf{v}_1, \ldots, \mathbf{v}_n)) = \begin{bmatrix} \lambda_1 - \lambda & & * \\ & \lambda_2 - \lambda & \\ & & \ddots \\ 0 & & & \lambda_n - \lambda \end{bmatrix}$$

Hence $T - \lambda I$ is not invertible if and only if λ equals one of the λ_j.
Hence λ is an eigenvalue of T if and only if λ equals one of the λ_j. □

Definition 1.2.56. A *diagonal matrix* is a square matrix that is 0 everywhere except possibly along diagonal.

Operator $T \in \mathcal{L}(\mathcal{V})$ has diagonal matrix $\text{diag}(\lambda_1, \ldots, \lambda_n)$ w.r.t. basis $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of $\mathcal{V}$ if and only if

$$\begin{aligned} T\mathbf{v}_1 &= \lambda_1 \mathbf{v}_1 \\ &\vdots \\ T\mathbf{v}_n &= \lambda_n \mathbf{v}_n \end{aligned}$$

This follows from definition of matrix of operator with respect to a basis,

If an operator has a diagonal matrix w.r.t. some basis, the entires along diagonal are precisely eigenvalues.

Proposition 1.2.57. *If $T \in \mathcal{L}(\mathcal{V})$ has $\dim(\mathcal{V})$ distinct eigenvalues, then T has a diagonal matrix with respect to some basis of $\mathcal{V}$.*

Proof. Suppose $T \in \mathcal{L}(\mathcal{V})$ has $\dim(\mathcal{V})$ distinct eigenvalues $\lambda_1, \ldots, \lambda_{\dim(\mathcal{V})}$. For each j, let $\mathbf{v} \in \mathcal{V}$ be nonzero eigenvector corresponding to the eigenvalue λ_j. As nonzero eigenvectors corresponding to distinct eigenvalues are linearly independent, $(\mathbf{v}_1, \ldots, \mathbf{v}_{\dim(\mathcal{V})})$ is linearly independent. A linearly independent list of $\dim(\mathcal{V})$ vectors in $\mathcal{V}$ is a basis of $\mathcal{V}$, hence $(\mathbf{v}_1, \ldots, \mathbf{v}_{\dim(\mathcal{V})})$ is a basis of $\mathcal{V}$. With respect to this basis, T is a diagonal matrix. □

Proposition 1.2.58. *Suppose $T \in \mathcal{L}(\mathcal{V})$. Let $\lambda_1, \ldots, \lambda_m$ denote distinct eigenvalues of T. Then the following are equivalent:*

i. T has a diagonal matrix with respect to some basis of $\mathcal{V}$;

ii. $\mathcal{V}$ has a basis consisting of eigenvectors of T;

iii. $\exists$ one-dimensional subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_n$ of $\mathcal{V}$, each invariant under T, such that

$$\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$$

iv. $\mathcal{V} = \text{null}(T - \lambda_1 I) \oplus \cdots \oplus \text{null}(T - \lambda_m I)$

v. $\dim(\mathcal{V}) = \dim(\text{null}(T - \lambda_1 I)) + \cdots + \dim(\text{null}(T - \lambda_m I))$

Proof. Already shown earlier that (i) and (ii) are equivalent.
$(ii) \Rightarrow (iii)$. Suppose (ii) holds, thus $\mathcal{V}$ has a basis $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ consisting of eigenvectors of T. For each j, let $\mathcal{U}_j = \text{span}(\mathbf{v}_j)$, and hence each $\mathcal{U}_j$ is a one-dimensional subspace of $\mathcal{V}$ that is invariant under T (as each $\mathbf{v}_j$ is an eigenvector of T). As $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$, each vector in $\mathcal{V}$ can be written uniquely as a linearly combination of $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$. Hence each vector in $\mathcal{V}$ can be written uniquely as a sum $\mathbf{u}_1 + \cdots + \mathbf{u}_n$, $\mathbf{u}_j \in \mathcal{U}_j$. Thus $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$, and (ii) implies (iii).
$(iii) \Rightarrow (ii)$. Suppose (iii) holds, hence there are one-dimensional subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_n$ of $\mathcal{V}$, each invariant under T, such that $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$. For each j, let $\mathbf{v}_j$ be nonzero vector in $\mathcal{U}_j$. Then each $\mathbf{v}_j$ is an eigenvector of T. As each vector in $\mathcal{V}$ can be written uniquely as $\mathbf{u}_1 + \cdots + \mathbf{u}_n, \mathbf{u}_j \in \mathcal{U}_j$, $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$.
Hence this shows so far $(i) \Leftrightarrow (ii) \Leftrightarrow (iii)$.
$(ii) \Rightarrow (iv)$. Suppose (ii) holds, then $\mathcal{V}$ has a basis consisting of eigenvectors of T, and hence every vector in $\mathcal{V}$ is a linear combination of eigenvectors of T. Thus

$$\mathcal{V} = \text{null}(T - \lambda_1 I) + \cdots + \text{null}(T - \lambda_m I)$$

Suppose $\mathbf{0} = \mathbf{u}_1 + \cdots + \mathbf{u}_m, \mathbf{u}_j \in \text{null}(T - \lambda_j I)$. As nonzero eigenvectors corresponding to distinct eigenvalues are linearly independent, this implies each $\mathbf{u}_j = \mathbf{0}$. This implies the sum is a direct sum.
$(iv) \Rightarrow (v)$. Suppose (iv) holds. By definition, each $\text{null}(T - \lambda_j I)$ is a subspace of $\mathcal{V}$. By definition of direct sum, each $\mathbf{V} \in \mathcal{V}$ can be uniquely expressed as $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_m, \mathbf{u}_j \in \text{null}(T - \lambda_j I)$.
The only way for each $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with each $\mathbf{u}_j \in \text{null}(T - \lambda_j I)$ is if each $\mathbf{u}_j = \mathbf{0}$. Hence these properties together imply $(iv) \Rightarrow (v)$.
$(v) \Rightarrow (ii)$. Suppose (v) holds. Choose a basis $\mathbf{v}_j \in \text{null}(T - \lambda_j I)$, and form a list $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of eigenvectors of T, where $n = \dim(\mathcal{V})$. Suppose $a_1 \mathbf{v}_1 + \cdots + a_n \mathbf{v}_n = \mathbf{0}$, where $a_1, \ldots, a_n \in \mathbb{F}$. For each $j = 1, \ldots, m$, let $\mathbf{u}_j$ be sum of all terms $a_k \mathbf{v}_k$ such that $\mathbf{v}_k \in \text{null}(T - \lambda_j I)$. Thus each $\mathbf{u}_j$ is an eigenvector of T with eigenvalue λ_j, and $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$. As nonzero eigenvectors corresponding to distinct eigenvalues are linearly independent, this implies each $\mathbf{u}_j = \mathbf{0}$. As each $\mathbf{u}_j$ is sum of $a_k \mathbf{v}_k$, where $\mathbf{v}_k$ were chosen to be a basis of $\text{null}(T - \lambda_j I)$, this implies all $a_k = 0$. Hence $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent and hence is a basis of $\mathcal{V}$. □

Next theorem shows that an invariant subspace of dimension 1 or 2 always exists.

Theorem 1.2.59. *Every operator on a finite-dimensional, nonzero, real vector space has an invariant subspace of dimension* 1 *or* 2.

Proof. Let $\mathcal{V}$ be a real vector space, $\dim(\mathcal{V}) = n > 0$, and $T \in \mathcal{L}(\mathcal{V})$. Choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$, then

$$(\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \dots, T^n\mathbf{v})$$

cannot be linearly independent as $\mathcal{V}$ has dimension n, and there are $n+1$ vectors.
Hence $\exists\ a_0, \dots a_n \in \mathbb{R}$ not all 0, such that

$$\mathbf{0} = a_0\mathbf{v} + a_1T\mathbf{v} + \cdots + a_nT^n\mathbf{v}$$

Let a be coefficients of a polynomial, then the factored form is

$$a_0 + a_1x + \cdots + a_nx^n = c(x - \lambda_1)\cdots(x - \lambda_m)(x^2\alpha_1x + \beta_1)\cdots(x^2 + \alpha_Mx + \beta_M)$$

where $c \neq 0 \in \mathbb{R}$, $\lambda_j, \alpha_j, \beta_j \in \mathbb{R}$, $m + M \geq 1$, and equation holds for all $x \in \mathbb{R}$. Then

$$\begin{aligned}\mathbf{0} &= a_0\mathbf{v} + a_1T\mathbf{v} + \cdots + a_nT^n\mathbf{v}\\ &= (a_0I + a_1T + \cdots + a_nT^n)\mathbf{v}\\ &= c(T - \lambda_1I)\cdots(T - \lambda_mI)(T^2 + \alpha_1T + \beta_1I)\cdots(T^2 + \alpha_MT + \beta_MI)\mathbf{v}\end{aligned}$$

This means $T - \lambda_jI$ is not injective for at least one j, or $(T^2 + \alpha_jT + \beta_jI)$ is not injective for at least one j. If $T - \lambda_jI$ is not injective for at least one j, then T has an eigenvalue, and hence a one-dimensional subspace.
If $(T^2 + \alpha_jT + \beta_jT)$ is not injective for some j, then there exists nonzero vector $\mathbf{u} \in \mathcal{V}$ such that

$$T^2\mathbf{u} + \alpha_jT\mathbf{u} + \beta_j\mathbf{u} = \mathbf{0}$$

Consider typical element of $\operatorname{span}(\mathbf{u}, T\mathbf{u})$ of form $a\mathbf{u} + bT\mathbf{u}$, where $a, b \in \mathbb{R}$, then

$$\begin{aligned}T(a\mathbf{u} + bT\mathbf{u}) &= aT\mathbf{u} + bT^2\mathbf{u}\\ &= aT\mathbf{u} - b\alpha_jT\mathbf{u} - b\beta_j\mathbf{u}\end{aligned}$$

last equality comes from solving the equality for $T^2\mathbf{u}$. This shows $T(a\mathbf{u}+bT\mathbf{u}) \in \operatorname{span}(\mathbf{u}, T\mathbf{u})$. hence $\operatorname{span}(\mathbf{u}, T\mathbf{u})$ is invariant under T. □

Definition 1.2.60. Let $\mathcal{U}$, $\mathcal{W}$ be subspaces of $\mathcal{V}$, where $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.
Let $\mathbf{u} \in \mathcal{U}, \mathbf{w} \in \mathcal{W}, \mathbf{v} \in \mathcal{V}$, and $\mathbf{v} = \mathbf{u} + \mathbf{w}$. Define *projection* $P_{\mathcal{U},\mathcal{W}} \in \mathcal{L}(\mathcal{V})$ by $P_{\mathcal{U},\mathcal{W}}\mathbf{v} = \mathbf{u}$. Note that

$$\begin{aligned}P_{\mathcal{W},\mathcal{U}}\mathbf{v} = \mathbf{w}, &\quad \mathbf{v} = P_{\mathcal{U},\mathcal{W}}\mathbf{v} + P_{\mathcal{W},\mathcal{U}}\mathbf{v}\\ \operatorname{range}(P_{\mathcal{U},\mathcal{W}}) = \mathcal{U}, &\quad \operatorname{null}(P_{\mathcal{U},\mathcal{W}}) = \mathcal{W}\end{aligned}$$

Theorem 1.2.61. *Every operator on an odd-dimensional real vector space has an eigenvalue.*

Proof. Suppose $\mathcal{V}$ is a real vector space with odd dimension.
Note that result holds for $\dim(\mathcal{V}) = 1$.
Suppose $\dim(\mathcal{V})$ is an odd number greater than 1. Suppose $T \in \mathcal{L}(\mathcal{V})$, to prove T has an eigenvalue. By Theorem 1.2.59, there is a two-dimensional subspace $\mathcal{U}$ of $\mathcal{V}$ that is invariant under T. Let $\mathcal{W}$ be any subspace of $\mathcal{V}$ such that $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then Theorem 1.2.59 guarantees that such a $\mathcal{W}$ exists.
As $\mathcal{W}$ has dimension 2 less than $\dim(\mathcal{V})$, apply induction hypothesis to $T|_{\mathcal{W}}$. However, $\mathcal{W}$ might not be invariant under T, which means $T_{\mathcal{W}}$ might not be an operator on $\mathcal{W}$.
Define $S \in \mathcal{L}(\mathcal{W})$ by $S\mathbf{w} = P_{\mathcal{W},\mathcal{U}}(T\mathbf{w})$ for $\mathbf{w} \in \mathcal{W}$. By induction, S has an eigenvalue λ.
Let $\mathbf{w} \in \mathcal{W}$ be nonzero eigenvector for S corresponding to the eigenvalue λ, thus $(S - \lambda I)\mathbf{w} = 0$. To look for an eigenvector of T in $\mathcal{U} + \operatorname{span}(\mathbf{w})$. Consider typical vector $\mathbf{u} + a\mathbf{w}$ in $\mathcal{U} = \operatorname{span}(\mathbf{w})$, where $\mathbf{u} \in \mathcal{U}$ and $a \in \mathbb{R}$.

$$\begin{aligned}(T - \lambda I)(\mathbf{u} + a\mathbf{w}) &= T\mathbf{u} - \lambda\mathbf{u} + a(T\mathbf{w} - \lambda\mathbf{w})\\ &= T\mathbf{u} - \lambda\mathbf{u} + a(P_{\mathcal{U},\mathcal{W}}(T\mathbf{w}) + P_{\mathcal{W},\mathcal{U}}(T\mathbf{w}) - \lambda\mathbf{w})\\ &= T\mathbf{u} - \lambda\mathbf{u} + a(P_{\mathcal{U},\mathcal{W}}(T\mathbf{w}) + S\mathbf{w} - \lambda\mathbf{w})\\ &= T\mathbf{u} - \lambda\mathbf{u} + aP_{\mathcal{U},\mathcal{W}}(T\mathbf{w})\end{aligned}$$

Note $T\mathbf{u} \in \mathcal{U}$ (as $\mathcal{U}$ is invariant under T), $\lambda\mathbf{u} \in \mathcal{U}$ (as $\mathbf{u} \in \mathcal{U}$), and $aP_{\mathcal{U},\mathcal{W}}(T\mathbf{w}) \in \mathcal{U}$ (by definition). Hence $T - \lambda I$ maps $\mathcal{U} + \text{span}(\mathbf{w})$ into $\mathcal{U}$. As $\mathcal{U} + \text{span}(\mathbf{w})$ has larger dimension than $\mathcal{U}$ this means $(T - \lambda I)|_{\mathcal{U}+\text{span}(\mathbf{w})}$ is not injective, and there exists a nonzero vector $\mathbf{v} \in \mathcal{U} + \text{span}(\mathbf{w})$ such that $(T - \lambda I)\mathbf{v} = \mathbf{0}$. Hence T has an eigenvalue as desired. □

1.2.4 Inner-Product Spaces

References

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